

EXPERIMENT NO. 9

AI-BASED TRAFFIC PREDICTION AND PERFORMANCE ANALYSIS USING PYTHON

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AIM

To evaluate a simple AI-based traffic-prediction model using Python in Google Colab by comparing actual and predicted traffic values collected over 7 hours and analyzing the prediction performance using suitable statistical measures and graphs.

OBJECTIVES

1. To analyze traffic data collected over 7 hours.
2. To consider actual and predicted traffic values.
3. To compare actual traffic with AI-predicted traffic.
4. To calculate prediction errors.
5. To evaluate the performance of the prediction model.
6. To calculate Mean Absolute Error (MAE).
7. To calculate Mean Squared Error (MSE).
8. To calculate Root Mean Squared Error (RMSE).
9. To visualize actual and predicted traffic values.
10. To demonstrate the use of Python for traffic prediction analysis.

SOFTWARE / APPARATUS REQUIRED

- Google Colab
- Python 3.x
- pandas
- numpy
- matplotlib

THEORY

Traffic prediction is an important application of Artificial Intelligence and Machine Learning in smart transportation systems. AI-based traffic prediction models estimate future traffic conditions using previously collected traffic data.

In this experiment, traffic data is considered for **7 hours**. The actual traffic values are compared with the corresponding predicted traffic values generated by a simple AI-based prediction model.

The traffic values are represented in **Mbps**.

Given Data

Hour	Actual Traffic (Mbps)	Predicted Traffic (Mbps)
1	42	44
2	49	47
3	57	59
4	64	62
5	72	74
6	79	77
7	85	82

The prediction error is calculated as:

$$\text{Error} = \text{Actual Traffic} - \text{Predicted Traffic}$$

The absolute error is:

$$\text{Absolute Error} = |\text{Actual} - \text{Predicted}|$$

Performance Metrics

Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum |\text{Actual} - \text{Predicted}|$$

Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{n} \sum (\text{Actual} - \text{Predicted})^2$$

Root Mean Squared Error (RMSE):

RMSE=MSE

Lower error values indicate better prediction performance.

DATA FLOW

Traffic Data



Actual Traffic Values



AI Prediction Model



Predicted Traffic Values



Compare Actual vs Predicted



Calculate Prediction Error



MAE / MSE / RMSE



Generate Graphs



Performance Analysis

ALGORITHM

1. Start the program.
2. Import pandas, numpy, and matplotlib.
3. Define the 7-hour traffic data.
4. Store the actual traffic values.
5. Store the predicted traffic values.
6. Create a pandas DataFrame.
7. Calculate the prediction error.
8. Calculate the absolute error.
9. Calculate squared error.
10. Calculate MAE.
11. Calculate MSE.
12. Calculate RMSE.
13. Display the actual and predicted traffic values.
14. Plot actual and predicted traffic using a line graph.
15. Plot prediction error using a bar graph.
16. Analyse the prediction performance.
17. Display the final result.
18. Stop the program.

GOOGLE COLAB CODE

```
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```

EXPERIMENT NO. 9

AI-BASED TRAFFIC PREDICTION AND PERFORMANCE ANALYSIS

GOOGLE COLAB VERSION

```
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```

```
import pandas as pd import numpy as np import matplotlib.pyplot  
as plt
```

```
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```

1. TRAFFIC DATA

```
=====
```

```
hours = [1, 2, 3, 4, 5, 6, 7]  
  
actual_traffic = [ 42, 49, 57, 64, 72, 79, 85 ]  
  
predicted_traffic = [ 44, 47, 59, 62, 74, 77, 82 ]
```

```
=====
```

2. CREATE DATAFRAME

```
=====
```

```
df = pd.DataFrame({  
  
    "Hour": hours,  
  
    "Actual Traffic (Mbps)": actual_traffic,
```

```
"Predicted Traffic (Mbps)": predicted_traffic

})
```

3. CALCULATE ERRORS

```
df["Error (Mbps)"] = ( df["Actual Traffic (Mbps)"] -
df["Predicted Traffic (Mbps)"] )

df["Absolute Error (Mbps)"] = ( df["Error (Mbps)"].abs() )

df["Squared Error"] = ( df["Error (Mbps)"] ** 2 )
```

4. PERFORMANCE METRICS

```
mae = df["Absolute Error (Mbps)"].mean()

mse = df["Squared Error"].mean()

rmse = np.sqrt(mse)
```

5. DISPLAY TRAFFIC DATA

```
print("=" * 70) print(" AI-BASED TRAFFIC PREDICTION ANALYSIS")
print("=" * 70)

print()

print("TRAFFIC DATA") print("-" * 70)
```

```
print( df.to_string( index=False ) )
```

```
=====
```

6. DISPLAY PERFORMANCE METRICS

```
=====
```

```
print() print("=" * 70) print(" PREDICTION PERFORMANCE")
```

```
print("=" * 70)
```

```
print()
```

```
print( "Mean Absolute Error (MAE) :", round(mae, 2), "Mbps" )
```

```
print( "Mean Squared Error (MSE) :", round(mse, 2) )
```

```
print( "Root Mean Squared Error :", round(rmse, 2), "Mbps" )
```

```
=====
```

7. GRAPH 1 - ACTUAL VS PREDICTED TRAFFIC

```
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```

```
plt.figure( figsize=(10, 5) )
```

```
plt.plot( df["Hour"], df["Actual Traffic (Mbps)"], marker="o",  
linewidth=2, label="Actual Traffic" )
```

```
plt.plot( df["Hour"], df["Predicted Traffic (Mbps)"],  
marker="s", linewidth=2, label="Predicted Traffic" )
```

```
plt.xlabel( "Hour" )
```

```
plt.ylabel( "Traffic (Mbps)" )
```

```
plt.title( "Actual vs Predicted Traffic" )
```

```
plt.xticks( hours )
```

```
plt.grid( True, linestyle="--", alpha=0.5 )
```

```
plt.legend()
```

```
plt.tight_layout()
```

```
plt.show()
```

```
=====
```

8. GRAPH 2 - PREDICTION ERROR

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```

```
plt.figure( figsize=(10, 5) )
```

```
plt.bar( df["Hour"], df["Error (Mbps)"] )
```

```
plt.axhline( 0, linewidth=1 )
```

```
plt.xlabel( "Hour" )
```

```
plt.ylabel( "Prediction Error (Mbps)" )
```

```
plt.title( "Traffic Prediction Error" )
```

```
plt.xticks( hours )
```

```
plt.grid( axis="y", linestyle="--", alpha=0.5 )
```

```
plt.tight_layout()
```

```
plt.show()
```

```
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```

9. SAVE DATA

```
=====
```

```
df.to_csv( "AI_Traffic_Prediction_Report.csv", index=False )
```

```
=====

10. FINAL ANALYSIS

=====
```

```
print() print("=" * 70) print(" FINAL ANALYSIS") print("=" * 70)
print()

if mae < 5: print( "Prediction Performance : GOOD" )
elif mae < 10: print( "Prediction Performance : MODERATE" )
else: print( "Prediction Performance : POOR" )

print()

print( "Maximum Absolute Error :", round( df["Absolute Error
(Mbps)"].max(), 2 ), "Mbps" )

print( "Minimum Absolute Error :", round( df["Absolute Error
(Mbps)"].min(), 2 ), "Mbps" )

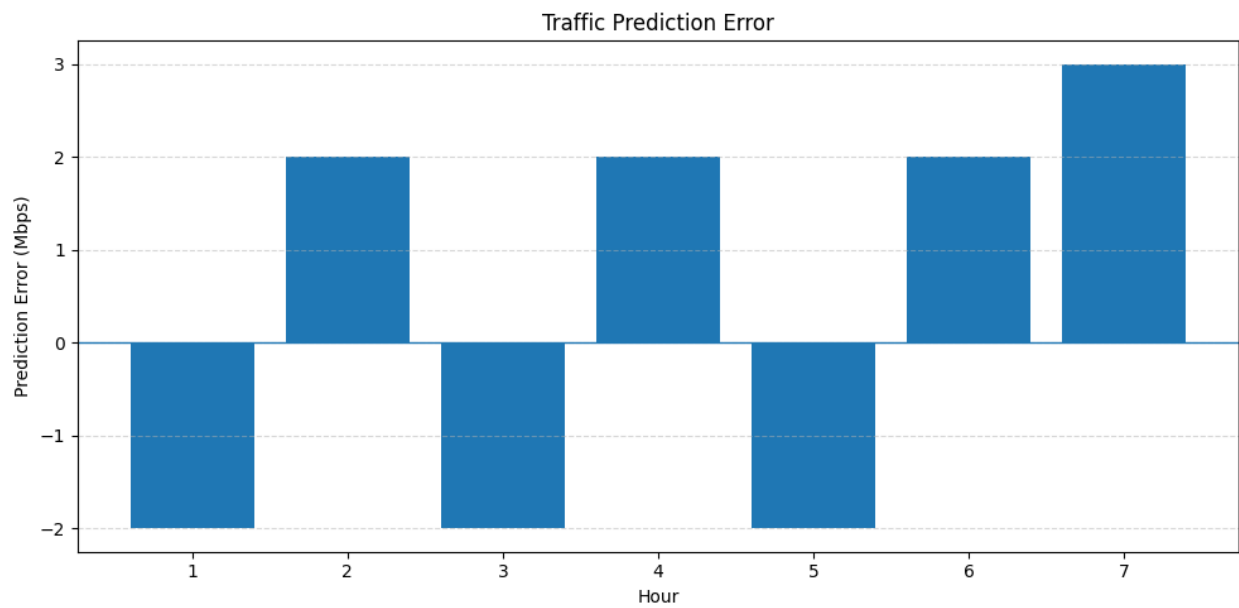
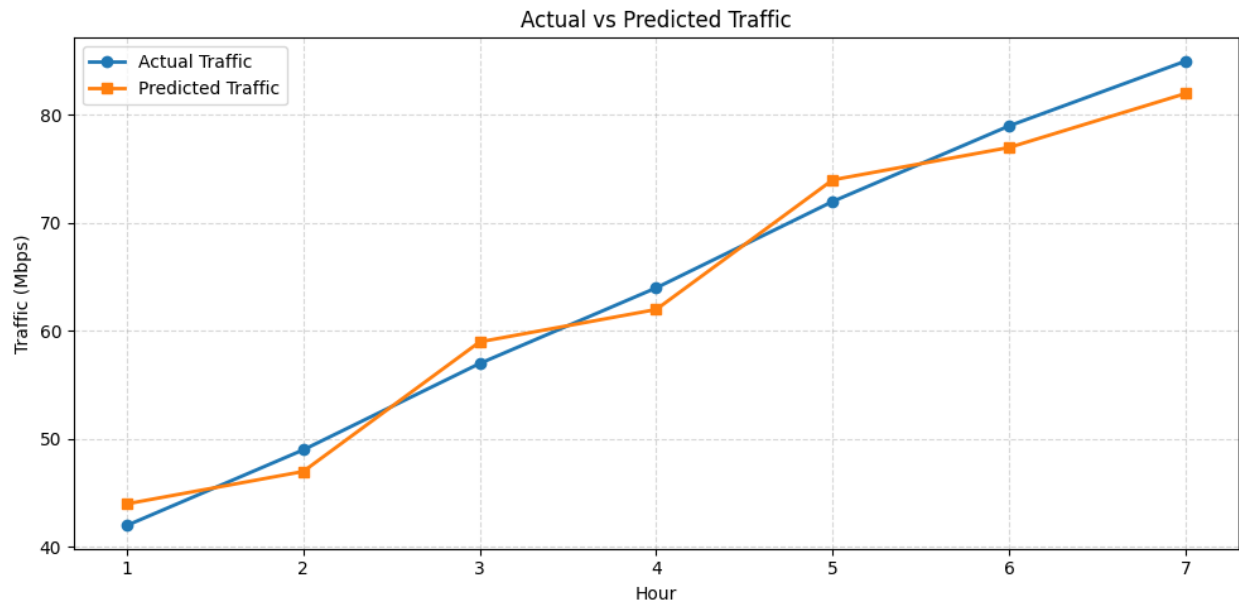
print()

print( "Report saved as: AI_Traffic_Prediction_Report.csv" )

print()

print("=" * 70) print( "AI-Based Traffic Prediction Analysis
Completed Successfully!" ) print("=" * 70)
```

OUTPUT



EXPECTED OUTPUT

Hour	Actual	Predicted	Error	Absolute Error
1	42	44	-2	2
2	49	47	2	2
3	57	59	-2	2
4	64	62	2	2
5	72	74	-2	2
6	79	77	2	2
7	85	82	3	3

OBSERVATION

S.No.	Parameter	Value
1	Observation Period	7 hours
2	Actual Traffic Range	42–85 Mbps
3	Predicted Traffic Range	44–82 Mbps
4	MAE	2.14 Mbps
5	MSE	4.71
6	RMSE	2.17 Mbps
7	Prediction Performance	Good
8	Visualization	Line and bar graphs

SAMPLE OUTPUT

```
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```

AI-BASED TRAFFIC PREDICTION ANALYSIS

```
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```

TRAFFIC DATA

```
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```

```
Hour   Actual Traffic (Mbps)  Predicted Traffic (Mbps)  Error
(Mbps) Absolute Error (Mbps)
```

1	42	44
-2	2	
2	49	47
2	2	
3	57	59
-2	2	
4	64	62
2	2	
5	72	74
-2	2	
6	79	77
2	2	
7	85	82
3	3	

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PREDICTION PERFORMANCE

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Mean Absolute Error (MAE) : 2.14 Mbps

Mean Squared Error (MSE) : 4.71

Root Mean Squared Error : 2.17 Mbps

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FINAL ANALYSIS

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Prediction Performance : GOOD

Maximum Absolute Error : 3 Mbps

Minimum Absolute Error : 2 Mbps

Report saved as: AI_Traffic_Prediction_Report.csv

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AI-Based Traffic Prediction Analysis Completed Successfully!

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RESULT

The AI-based traffic-prediction model was successfully evaluated using Python in Google Colab. The actual and predicted traffic values for seven hours were compared, and the prediction errors were calculated.